Chapter 2 Answer Key

Introduction to Statistics — MATH& 146

Section 2.1 Exercises

2.1.1. Bag: 3 red, 2 blue, 1 green marble. Total = 6.

x	Color	$P(X = x)$
1	Red	$3/6 = 1/2$
2	Blue	$2/6 = 1/3$
3	Green	$1/6$

Verification: $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = \frac{3}{6} + \frac{2}{6} + \frac{1}{6} = \frac{6}{6} = 1.$ ✓

2.1.2. **Not a valid distribution.** Two rules are violated:

- Rule 1 fails: $P(X = 3) = -0.1 < 0$. Probabilities cannot be negative.
- (Rule 2 technically holds: $0.2 + 0.4 + 0.5 + (-0.1) = 1.0$, but the negative entry makes Rule 1 sufficient grounds for rejection.)

2.1.3. Four sectors with probabilities 0.3, 0.3, 0.2, 0.2. The probability histogram has four bars of equal width at sectors 1–4, with heights 0.3, 0.3, 0.2, and 0.2 respectively. The two taller bars are equal and the two shorter bars are equal; total area $= 0.3 + 0.3 + 0.2 + 0.2 = 1.0$.

Section 2.2 Exercises

2.2.1. Game costs \$2. Roll a die: roll a 6 $\Rightarrow$ win \$10 (net = +\$8); otherwise $\Rightarrow$ win nothing (net = −\$2).

Outcome	x (net gain)	$P(X = x)$
Roll a 6	$+8$	$1/6$
Roll 1–5	-2	$5/6$

$$\mu = 8 \cdot \tfrac{1}{6} + (-2) \cdot \tfrac{5}{6} = \frac{8 - 10}{6} = -\frac{2}{6} = -\frac{1}{3} \approx -\$0.33.$$

The game is **not fair**: the player loses about 33 cents per play on average. A fair game requires $\mu = 0$.

2.2.2. Using $\mu = \$150$ from Example 2.2.2:

x	$P(X=x)$	$x-\mu$	$(x-\mu)^2$	$(x-\mu)^2 \cdot P$
0	0.97	-150	22,500	21,825
5000	0.03	$+4850$	23,522,500	705,675
			$\sigma^2 =$	727,500

The correct value of σ^2 is **727,500**. **Note:** the target value of 14,550 stated in the problem appears to be a typographical error in the exercise. No combination of values consistent with Example 2.2.2 produces 14,550.

2.2.3. There is **no single correct answer** — this is a question about risk preference, not arithmetic.

- Both investments have the same expected return $\mu = \$500$.
- Investment A ($\sigma = \$50$) is highly predictable: nearly all outcomes will be close to \$500.
- Investment B ($\sigma = \$400$) is volatile: outcomes range widely, offering the possibility of much higher *or* much lower returns.
- A risk-averse investor (e.g., near retirement, needing stable income) would prefer A. A risk-tolerant investor seeking high upside might prefer B. Standard deviation quantifies the tradeoff but does not dictate the choice.

Section 2.3 Exercises

$p = 0.08$, $q = 1 - p = 0.92$. X = number of parts inspected until first defect.

2.3.1. (a) Individual probabilities:

$$P(X=1) = (0.92)^0 \cdot 0.08 = \mathbf{0.0800}$$
$$P(X=2) = (0.92)^1 \cdot 0.08 = 0.92 \times 0.08 = \mathbf{0.0736}$$
$$P(X=3) = (0.92)^2 \cdot 0.08 = 0.8464 \times 0.08 \approx \mathbf{0.0677}$$

(b) Mean and standard deviation:

$$\mu = \frac{1}{p} = \frac{1}{0.08} = \mathbf{12.5} \text{ parts.} \qquad \sigma = \frac{\sqrt{1-p}}{p} = \frac{\sqrt{0.92}}{0.08} = \frac{0.9592}{0.08} \approx \mathbf{11.99} \text{ parts.}$$

The inspector expects to check about 12.5 parts before finding a defect, with substantial variability.

(c) The first six bars have heights (rounded to four decimal places):

$$k=1: 0.0800; \quad k=2: 0.0736; \quad k=3: 0.0677; \quad k=4: 0.0623; \quad k=5: 0.0573; \quad k=6: 0.0527.$$

The histogram is sharply decreasing from left to right, with the tallest bar at $k=1$ and a very long right tail (small p means many trials may be needed).

2.3.2. $p = 1/4 = 0.25$ (one correct choice out of four). First correct answer on question 5:

$$P(X=5) = (0.75)^4 \cdot 0.25 = 0.31640625 \times 0.25 \approx \mathbf{0.0791}.$$

About a 7.9% chance.

2.3.3. We must show $\displaystyle\sum_{k=1}^{\infty}(1-p)^{k-1} \cdot p = 1$.

Proof. Let $q = 1 - p$. Then

$$\sum_{k=1}^{\infty} q^{k-1} \cdot p = p\sum_{k=1}^{\infty} q^{k-1} = p\sum_{j=0}^{\infty} q^j \quad (j = k-1) = p \cdot \frac{1}{1-q} = p \cdot \frac{1}{p} = 1,$$

where the geometric series converges because $|q| = |1-p| < 1$ for $0 < p < 1$. □

Section 2.4 Exercises

2.4.1. $n = 20$, $p = 0.05$, $X =$ number of defective bolts.

(a) **Approximately binomial:** (1) Fixed number of trials ($n = 20$). (2) Each bolt is defective (success) or not (failure). (3) Constant defect rate $p = 0.05$ across bolts. (4) Bolts are independent, or approximately so: the sample of 20 is almost certainly less than 10% of total production, so sampling without replacement introduces negligible dependence.

(b)

$$P(X=0) = \binom{20}{0}(0.05)^0(0.95)^{20} = (0.95)^{20} \approx \mathbf{0.3585}$$
$$P(X=1) = \binom{20}{1}(0.05)^1(0.95)^{19} = 20(0.05)(0.95)^{19} \approx 20(0.05)(0.3774) \approx \mathbf{0.3774}$$

(c)

$$\mu = np = 20(0.05) = \mathbf{1} \text{ defective bolt (expected).}$$
$$\sigma = \sqrt{np(1-p)} = \sqrt{20(0.05)(0.95)} = \sqrt{0.95} \approx \mathbf{0.975}.$$

(d) The histogram is strongly **skewed right**. With $p = 0.05$ and $n = 20$, zero or one defective is overwhelmingly likely (together accounting for about 73.6% of the probability). Bars decrease rapidly as k increases, with the bar at $k = 0$ being the tallest.

2.4.2. $n = 15$, $p = 0.20$, $X =$ number of correct answers.

(a) $\mu = np = 15(0.20) = \mathbf{3}$ correct answers expected.

(b) $P(X \geq 5) = 1 - P(X \leq 4)$. Computing each term:

$$P(X=0) = (0.80)^{15} \approx 0.0352$$
$$P(X=1) = \tbinom{15}{1}(0.20)(0.80)^{14} \approx 0.1319$$
$$P(X=2) = \tbinom{15}{2}(0.20)^2(0.80)^{13} \approx 0.2309$$
$$P(X=3) = \tbinom{15}{3}(0.20)^3(0.80)^{12} \approx 0.2501$$
$$P(X=4) = \tbinom{15}{4}(0.20)^4(0.80)^{11} \approx 0.1876$$

$$P(X \le 4) \approx 0.8357 \implies P(X \ge 5) \approx \mathbf{0.1643}.$$

There is about a 16.4% chance of getting 5 or more correct by guessing.

(c) **Passing by guessing is not realistic.** The expected score is only 3 out of 15, and $\sigma = \sqrt{15(0.20)(0.80)} \approx 1.55$. A score of 9 is $(9-3)/1.55 \approx 3.9$ standard deviations above the mean — an extremely rare outcome. Numerically, $P(X \ge 9) < 0.001$; the chance of passing is less than 0.1%.

2.4.3. As n increases from 5 to 10 to 20 (with $p = 0.5$):

- The histogram remains perfectly symmetric about $k = n/2$ (since $p = 0.5$).
- The bars become narrower and more numerous, and the outline of the histogram smooths out.
- The shape increasingly resembles a **bell curve** (the normal distribution). This is the first visible hint of the Central Limit Theorem.

Chapter 2 Review

1. Carnival game: costs \$3. Draw from a 52-card deck.

- Ace (4 cards): win \$20, net = +\$17, $P = 4/52 = 1/13$.
- Face card, J/Q/K (12 cards): win \$8, net = +\$5, $P = 12/52 = 3/13$.
- Other (36 cards): win \$0, net = −\$3, $P = 36/52 = 9/13$.

(a) Distribution table:

x (net gain)	$P(X = x)$
+17	1/13
+5	3/13
−3	9/13

(b)

$$\mu = 17\cdot\tfrac{1}{13} + 5\cdot\tfrac{3}{13} + (-3)\cdot\tfrac{9}{13} = \frac{17 + 15 - 27}{13} = \frac{5}{13} \approx \$\mathbf{0.38}.$$

The game is **slightly in the player's favor**: the expected net gain is about 38 cents per play.

(c) Variance (using $\mu = 5/13$):

x	P	$x - \mu$	$(x-\mu)^2$	$(x-\mu)^2 \cdot P$
+17	1/13	216/13	46,656/169	46,656/2,197
+5	3/13	60/13	3,600/169	10,800/2,197
−3	9/13	−44/13	1,936/169	17,424/2,197
			$\sigma^2 =$	74,880/2,197 ≈ 34.08

$$\sigma^2 \approx 34.08, \qquad \sigma \approx \$\mathbf{5.84}.$$

2. Circuit boards, $p = 0.04$ per board, independent.

(a) Geometric: first defect on the 5th board inspected.

$$P(X = 5) = (0.96)^4 \cdot 0.04 = 0.84935 \times 0.04 \approx \mathbf{0.0340}.$$

(b) $\mu = 1/p = 1/0.04 = \mathbf{25}$ boards. The inspector expects to check 25 boards on average before finding a defect.

(c) Inspecting exactly 25 boards and *counting* defectives is a **binomial** situation: the number of trials is fixed at $n = 25$, and X counts how many of those 25 boards are defective. (Geometric asks how long until the *first* success; binomial asks how many successes in a *fixed* number of trials.)

(d) $X \sim \text{Bin}(25,\ 0.04)$:

$$P(X = 0) = (0.96)^{25} \approx \mathbf{0.3604}$$

$$P(X = 1) = \binom{25}{1}(0.04)^1(0.96)^{24} = 25(0.04)(0.96)^{24} \approx 25(0.04)(0.3754) \approx \mathbf{0.3754}$$

$$\mu = np = 25(0.04) = \mathbf{1} \text{ defective board (expected).}$$

3. $\text{Bin}(10, 0.1)$: skewed right. Most probability is concentrated at $k = 0$ and $k = 1$; the histogram has a tall bar at $k = 0$ and bars that decrease rapidly, with a long right tail.

$\text{Bin}(10, 0.9)$: skewed left. Most probability is concentrated at $k = 9$ and $k = 10$; the histogram is the mirror image of the $p = 0.1$ histogram.

They are related by symmetry: if $X \sim \text{Bin}(10, 0.1)$, then $Y = 10 - X \sim \text{Bin}(10, 0.9)$, so $P(\text{Bin}(10, 0.1) = k) = P(\text{Bin}(10, 0.9) = 10 - k)$ for every k. High-success outcomes under $p = 0.9$ correspond exactly to low-success outcomes under $p = 0.1$.

4. Model response (writing prompt):

In a discrete probability histogram, each bar has width 1, so the bar's area equals its height — which equals its probability. Probability *is* area, not merely proportional to it. This matters because in Chapter 3 we will let the number of bars grow without bound, making them infinitesimally thin; the bars' tops trace a smooth curve, and probability becomes the area *under* that curve over any interval. The discrete and continuous frameworks are the same idea, expressed at different levels of resolution.